Experimental Results Summary

This document summarizes the performance comparison between traditional Encoder models (Legal-BERT) and the proposed Decoder-based Embedding models (Qwen-0.6B), across different training regimes on the `ecthr_a` dataset.

Performance Table (Test Set)

Category	Model	Method	Micro-F1	Macro-F1	Notes
Baseline	Legal-BERT (Base)	Zero-shot (CLS)	0.248	0.112	Limited by 512 context window.
Baseline	Qwen-Embedding-0.6B	Zero-shot (Last Token)	0.196	0.179	Raw embedding without adaptation.
Supervised	Qwen-Embedding-0.6B	LoRA Finetuned (Head)	0.563	0.413	Standard classification head.
Supervised	**Legal-BERT**	**Full Finetuned**	**0.692**	**0.600**	Strongest standard baseline.
Supervised	Legal-BERT (Hierarchical)	Full Finetuned	0.756	0.678	SOTA architecture for long docs.
RAG (Ours)	**Qwen-Embedding-0.6B**	**Training-Free k-NN**	**0.678**	**0.581**	**Best Efficiency / Performance**

Category	Model	Method	Micro-F1	Macro-F1	Notes
					Ratio. $(k = 10, \tau = 0.4)$

Key Findings

1. **RAG Efficacy:** Our **Training-Free RAG** approach (0.678 Micro-F1) significantly outperforms the standard Zero-shot baselines and even the LoRA-finetuned classification head (0.563).
2. **Competitiveness:** It nearly matches the performance of the fully fine-tuned **Legal-BERT** (0.692), despite requiring **no gradient updates** or training compute.
3. **Parameter Sensitivity:** Optimizing the k-NN threshold (from 0.5 to 0.4) yielded a ~4% gain in Macro-F1, highlighting the importance of calibration in retrieval-based classification.

Next Steps

- **LoRA-Adapted RAG:** Evaluate the k-NN performance using the Qwen embedding model fine-tuned via Contrastive Loss (in progress). This is expected to bridge the gap to the Hierarchical SOTA.